

7.02 Graphs and Networks

7.02a	Terminology and notation	a) Understand the meaning of the terms “vertex” (or “node”) and “arc” (or “edge”). <i>Includes the concept of the “degree” of a vertex as the number of arcs “incident” to the vertex.</i> <i>Includes the term “adjacent” for pairs of vertices or edges.</i>	
7.02b		b) Understand the meaning of the terms “tree”, “simple”, “connected” and “simply connected” as they refer to graphs. <i>Includes understanding and using the restrictions on the vertex degrees implied by these conditions.</i>	
7.02c		c) Understand the meaning of the terms “walk”, “trail”, “path”, “cycle” and “route”. <i>A “walk” is a set of arcs where the end vertex of one is the start vertex of the next.</i> <i>A “trail” is a walk in which no arcs are repeated.</i> <i>A “path” is a trail in which no nodes are repeated.</i> <i>A “cycle” is a closed path.</i> <i>A “route” can be a walk, a trail or a path, or may be a closed</i>	

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
7.02d	Complete graphs	d) Understand and be able to use the term “complete” and the notation K_n for a complete graph on n vertices. <i>Includes knowing that K_n has $\frac{1}{2}n(n-1)$ arcs.</i>	
7.02e 7.02f	Bipartite graphs	e) Understand and be able to use bipartite graphs and the notation $K_{m,n}$ for a complete bipartite graph connecting m vertices to n vertices. <i>Includes knowing that $K_{m,n}$ has mn arcs.</i>	f) Use a colouring argument to show that a given graph is, or is not, bipartite.
7.02g	Eulerian graphs	g) Use the degrees of vertices to determine whether a given graph is Eulerian, semi-Eulerian or neither. Understand what these terms mean in terms of traversing the graph.	
7.02h 7.02i	Hamiltonian graphs		h) Understand and be able to use the definition of a Hamiltonian path, a Hamiltonian cycle and a Hamiltonian graph. i) Know and use Ore’s theorem. <i>i.e. For a simple graph G with $n \geq 3$ vertices, if $\deg v + \deg w \geq n$ for every pair of distinct non-adjacent vertices v and w, then G is Hamiltonian.</i> <i>Includes understanding that Ore’s theorem gives a sufficient but not necessary condition for a graph to be Hamiltonian.</i>

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7.02j	Isomorphism	j) Understand what it means to say that two graphs are isomorphic. Construct an isomorphism either by a reasoned argument or by explicit labelling of vertices. <i>Includes understanding that having the same degree sequence (ordered list of vertex degrees) is necessary but not sufficient to show isomorphism.</i> <i>Includes the term “non-isomorphic”.</i>	
7.02k	Digraphs	k) Understand and be able to use digraphs. <i>Includes the terms “indegree” and “outdegree”.</i>	
7.02l 7.02m 7.02n 7.02o	Planar graphs		l) Understand and be able to apply the concepts of planarity, subdivision and contraction. <i>i.e. Subdivision is inserting a vertex of degree 2 into an arc.</i> <i>Contraction is contracting two vertices into one so that any arc incident with the original two vertices is incident with the contracted vertex.</i> <i>Includes the notation (AB) for the vertex created by the contraction of the arc AB.</i> <i>Includes drawing planar representations.</i> m) Know, understand and use Euler’s formula $V + R = E + 2$. n) Know and use Kuratowski’s theorem. <i>i.e. A finite graph is planar if and only if it does not contain a subgraph that is a subdivision of K_5 or $K_{3,3}$.</i> o) Understand and be able to use the concept of thickness. <i>[Calculation of thickness > 2 is excluded.]</i>

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7.02p	Using graphs and networks	p) Understand that a network is a weighted graph. Use graphs and networks to model the connections between objects. <i>Graphs and networks may be directed or undirected.</i>	
7.02q		q) Use an adjacency matrix representation of a graph and a weighted matrix representation of a network.	
7.02r		r) Be able to model problems using graphs or networks, and solve them.	